

There and Back Are Not the Same: Finsler Flow Matching for Asymmetric Robot Planning

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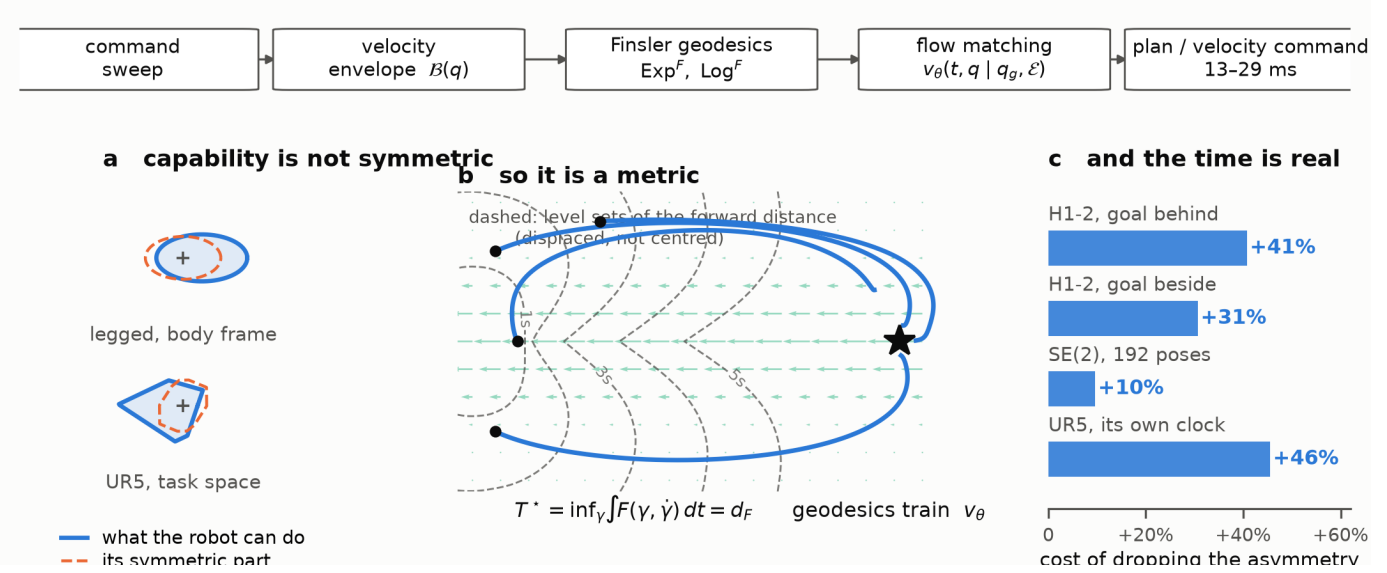


Fig. 1. There and back are not the same. A robot’s velocity envelope is measured, not designed, and it is not centrally symmetric; its gauge is therefore a non-reversible Finsler metric whose forward distance is exactly the minimum travel time, and flow matching amortizes that metric’s geodesics into a field that answers a planning query in one ODE solve. (a) The envelope of two very different bodies against the symmetric part a reversible metric is confined to. (b) Level sets of the forward distance in a drift field are displaced rather than centred, and the plans are integral curves of the learned field. (c) Time a planner loses by using the symmetric part, on a simulated humanoid, a UR5 and 192 held-out SE(2) poses.

Abstract—Robot motion capabilities are direction dependent: walking forward and backward, or moving with and against a flow, take substantially different times, and the Euclidean and Riemannian metrics that geometric planners rely on are reversible and cannot express that. We introduce Finsler flow matching, which reads the gauge of a robot’s measured velocity envelope as a non-reversible Finsler metric, whose forward distance is exactly the minimum travel time, and uses its geodesics as the conditional paths of a flow-matching model. Two properties of non-reversible geometry have to be respected, and we give both: the direction towards a goal is not the negative gradient of the distance, and the loss weight must not depend on the sampled endpoint. Across SE(2) pose planning, spatially varying drift fields and obstacle environments, the learned planner matches exact time-optimal geodesics to within 0.3%, where Euclidean and symmetric Riemannian conditional paths cost up to $2.6\times$ and $1.13\times$. Conditioned on the envelope, one field also plans for capability it has never seen, at 13–29ms per query, where a grid solver rebuilds its graph for every new metric. A simulated Unitree H1-2 humanoid reaches goals behind and beside it up to 29% faster when its planner knows the asymmetry. **On hardware, one envelope-conditioned network plans for a Unitree Go2 under three capability conditions and for an H1-2, from measured envelopes, and adapts to a payload attached mid-run at the next control step.**

I. INTRODUCTION

Robot motion capabilities are rarely symmetric. A legged robot walks forward faster than backward and sidesteps slower still, a vehicle turns more readily to one side, an arm that may approach a workpiece only slowly enough to keep the transferred energy within a safety limit can retract from it at full speed, and wind or current assists motion in one direction while opposing it in the other. The set of velocities a robot can sustain at a configuration, its velocity envelope, is therefore a convex body that is not centrally symmetric, and the minimum time to reach a goal depends on the direction of travel.

Geometric planners and geometric generative policies encode motion cost with a Euclidean or Riemannian metric. Both are reversible, $\|v\|_G = \|-v\|_G$, so the local cost of moving backward equals that of moving forward. A reversible metric can express anisotropy (moving sideways is expensive) but not asymmetry (moving backward is more expensive than moving forward). A planner built on the symmetric part of an asymmetric envelope backs up at the wrong speed, approaches goals from the wrong side, and misjudges travel time.

Finsler geometry replaces the quadratic norm by a direction-dependent norm $F(q, v)$ that is positively homogeneous but not necessarily even. The gauge function of the velocity

envelope is such a norm, and its forward distance is exactly the minimum travel time, so the planning geometry is derived from the measured envelope rather than designed. Its geodesics, however, are as expensive to compute as any time-optimal control solution: a boundary value problem per query, or a graph search whose memory grows exponentially with dimension. We amortize them with flow matching, which is not a matter of swapping the metric in an existing method. Forward and reverse geodesics differ, so the log map *to* a goal cannot be obtained by negating the log map *from* it; the fundamental tensor depends on velocity, so the natural metric-weighted regression loss weights each training sample by its own target and biases the learned flow; and the gradient of the distance function is no longer the negative of the geodesic direction.

Related work. Riemannian motion policies and geometric fabrics shape motion by pulling back Riemannian metrics through task maps [17, 7], and Ratliff et al. [18] generalized fabrics to Finsler energies for second-order behavior design; learned Riemannian manifolds yield geodesic motion skills [3]. None of these plan time-optimal paths under an asymmetric velocity envelope. Time-optimal navigation in a flow is Zermelo’s problem [25], whose solutions are geodesics of a Randers metric [2] and are computed per query by anisotropic fast marching [16] or level-set methods [14]. Flow matching [13, 21] and its Riemannian extension [6] regress a vector field from conditional paths, metric flow matching [12] learns the Riemannian metric that shapes those paths, and Riemannian flow matching policies [4], diffusion policies and diffusion planners [8, 10, 5] generate robot motion on symmetric geometries. Streaming flow policies [11] treat the flow trajectory as the action trajectory, a viewpoint we adopt for planning. All of these assume a reversible metric, whereas robot velocity envelopes and drift fields are generally non-reversible; quasimetric reinforcement learning [23] learns asymmetric value functions but provides no geometry with geodesics to plan along. Traversability estimation [24, 9] and learned locomotion controllers with velocity-command interfaces [19, 15] characterize what a robot can do; we identify the closed-loop reachable velocity envelope of such a controller and use it as the planning geometry, without identifying the full dynamics.

Contributions. (1) We represent asymmetric robot capability as the Finsler metric induced by the velocity envelope, so that the minimum-time planning geometry is measured rather than designed, with an interpretable Randers (Zermelo navigation) family and a structured learnable family of smooth, strongly convex metrics fitted from rollouts. (2) We construct flow-matching conditional paths and target fields from Finsler geodesics, taking the target from the forward distance through the Legendre transform and extending it to left-invariant metrics on SE(2) for body-frame envelopes. Two properties of non-reversible geometry have to be respected for this to be correct, and we give both: the direction towards a goal is not the negative gradient of the distance, and the loss weight must not depend on the sampled endpoint. (3) We

amortize over the capability envelope, not only over starts and goals. One conditioned field answers a start, a goal and an envelope it has never seen in milliseconds, where a grid solver must rebuild its graph for every new metric. We verify the resulting plans against exact time-optimal geodesics rather than a discretized reference, and validate obstacles, extrapolation to unseen asymmetry, envelopes learned from rollouts, closed-loop disturbance recovery, and execution on a simulated humanoid **and on two robots whose envelopes are measured, including a capability that changes mid-run.**

II. FINSLER FLOW PLANNING

A. Asymmetric capability geometry

Let $\mathcal{M}$ be the configuration manifold ($\mathbb{R}^2$ or SE(2) here) and $\mathcal{B}(q) \subset T_q\mathcal{M}$ the set of velocities the robot can sustain at q : a compact convex body with the origin in its interior. Its gauge function

$$F(q, v) = \inf\{\alpha > 0 : v/\alpha \in \mathcal{B}(q)\} \quad (1)$$

is positively one-homogeneous and convex, and we assume it is smooth away from $v = 0$ and strongly convex, so that F is a Finsler metric [1]. The minimum time from q_s to q_g is the forward Finsler distance

$$T^*(q_s, q_g) = \inf_{\gamma} \int_0^1 F(\gamma(t), \dot{\gamma}(t)) dt = d_F(q_s, q_g), \quad (2)$$

since a curve of Finsler length ℓ traversed on the envelope boundary $F = 1$ takes exactly time ℓ and no feasible traversal is faster; the minimum-time plan is a forward geodesic executed at unit Finsler speed. The single property that separates this geometry from every Riemannian one is

$$F(q, v) \neq F(q, -v), \quad \text{whereas} \quad \|v\|_G = \|-v\|_G. \quad (3)$$

Two families of F are used. The *Randers* metric $F(q, v) = \sqrt{v^\top A(q)v} + b(q)^\top v$ with $\|b\|_{A^{-1}} < 1$ is equivalent to Zermelo navigation [2]: a body whose own velocities fill the unit ball of a Riemannian metric h , carried by a drift W with $\|W\|_h < 1$, has the displaced ellipsoidal envelope $\{v : \|v - W\|_h \leq 1\}$, and its dual norm $F^*(q, \xi) = \xi^\top W + \|\xi\|_{h^{-1}}$ is closed form. The *p-sum* family $F(q, v) = (\sum_{k=1}^K \|L_k(q)v\|^p)^{1/p} + b(q)^\top v$, with b bounded relative to L_1 , is positive and one-homogeneous for any parameters, is strongly convex once a small regularizing ellipsoid is included, contains Randers ($K = 1$, $p = 1$), and approaches box-like envelopes for large p ; its parameters, including p , are fitted to velocity observations by least squares on the indicatrix $\{F = 1\}$. A legged robot’s envelope is specified in its body frame by forward, backward, lateral and yaw speeds, and the left-invariant metric $F(q, \dot{q}) = F_{\text{body}}(\Lambda(\theta)^{-1}\dot{q})$ on SE(2), with $\Lambda(\theta)$ the body-to-world map, carries it to every pose. Geodesics are integral curves of the Hamiltonian $\frac{1}{2}F^*(q, p)^2$ in general and of the Euler–Poincaré equations in body momentum on SE(2); the closed forms are given in the supplementary material.

The construction is not specific to legged robots. A manipulator’s achievable end-effector velocity set $J(q)\mathcal{Q}$ is

anisotropic and configuration dependent, and it becomes non-reversible as soon as a limit applies to one direction only. A safe-approach bound keeps the transferred energy $\frac{1}{2}m_{\text{eff}}(u)v^2$ below a threshold along the approach direction while retraction is unconstrained, where $m_{\text{eff}}(u) = (u^\top \Lambda^{-1} u)^{-1}$ is the effective mass of the operational-space inertia Λ . Effective mass is quadratic in u and therefore reversible by itself; the asymmetry comes from the constraint being one sided, and from gravity, which displaces the achievable set exactly as a drift does. Grasping an object changes Λ , so a field conditioned on the envelope plans for whatever is in the gripper without retraining, which is the manipulation counterpart of the payload conditioning of Sec. III-C.

B. Finsler flow matching

The fundamental tensor $g_v = \frac{1}{2}\partial_{vv}^2 F^2$ depends on v , and the Legendre transform $\mathcal{L}(v) = g_v(v, \cdot) = F \partial_v F$ is one-homogeneous but nonlinear, with inverse $\mathcal{L}^{-1}(\xi) = F^*(\xi) \partial_\xi F^*(\xi)$. The Finsler gradient $\nabla^F f = \mathcal{L}^{-1}(df)$ therefore satisfies $\nabla^F(-f) \neq -\nabla^F f$, and the reverse metric $\bar{F}(q, v) = F(q, -v)$ has $d_{\bar{F}}(x, y) = d_F(y, x)$. Flow matching needs the direction *towards* a goal, which for a non-reversible metric is not the negative gradient of the distance.

Proposition 1 (Forward geodesic velocity): Let $\rho(x) = d_F(x, x_1)$. Wherever ρ is differentiable, the unit-speed minimizing geodesic from x to x_1 starts with velocity

$$v(x) = \mathcal{L}_x^{-1}(-d\rho_x), \quad F(v) = 1, \quad d\rho(v) = -1. \quad (4)$$

The proof (supplementary) reads ρ as the distance *from* x_1 in the reverse metric, whose gradient is the arrival velocity of the reverse geodesic, and uses $\bar{g}_u = g_{-u}$. Shooting along $-\nabla^F \rho$ instead misses x_1 even for a constant Randers metric, so the premetric recipe of Riemannian flow matching does not transfer verbatim.

Given a start q_0 and a goal q_1 , the conditional path is the minimizing geodesic at constant Finsler speed,

$$\begin{aligned} q_t &= \text{Exp}_{q_0}^F(t \text{Log}_{q_0}^F(q_1)), \\ u_t(q | q_1) &= \dot{q}_t = \frac{\rho(q)}{1-t} \mathcal{L}_q^{-1}(-d\rho_q), \end{aligned} \quad (5)$$

which satisfies $\rho(q_t) = (1-t)\rho(q_0)$, the linear schedule of Riemannian flow matching [6]; a general decreasing schedule $\kappa(t)$ replaces $1/(1-t)$ by $-\dot{\kappa}/\kappa$. For a constant envelope $\rho(q) = F(q_1 - q)$ and (5) reduces to $u_t = (q_1 - q)/(1-t)$, so Euclidean optimal-transport flow matching [21] is the special case of a constant envelope, and direction-dependent paths arise exactly when the envelope varies over $\mathcal{M}$ or the metric lives on $\text{SE}(2)$. When only the goal distribution is prescribed, $q_0 = \text{Exp}_{q_1}^{\bar{F}}(w)$ for a sampled w gives q_t and u_t in closed form without any boundary value problem, a Finsler noising process; for planning, where starts and goals are both prescribed, we solve the geodesic boundary value problem offline.

The training objective is the metric-weighted conditional flow-matching loss

$$\mathcal{L} = \mathbb{E}_{t, (q_0, q_1)} \left[\|v_\theta(t, q_t | q_g, \mathcal{E}) - u_t\|_{M(t, q_t)}^2 \right]. \quad (6)$$

Remark 1 (Admissible loss weights): If the weight M is positive definite and depends only on (t, q_t) , the population minimizer of (6) is the marginal probability-flow field $\mathbb{E}[u_t | q_t]$, for every such M . If M depends on the conditional velocity u_t , in particular $M = g_{u_t}$, the minimizer is $(\mathbb{E}[M | q_t])^{-1} \mathbb{E}[M u_t | q_t]$, which differs from the marginal field whenever M and u_t are correlated given q_t .

The Riemannian choice $M = g(q_t)$ is admissible because $g_v \equiv g(q)$; the Finsler analogue g_{u_t} is not, and the supplementary material shows on a three-mode target that it collapses the generated mass onto the slow direction of a constant Randers metric. We therefore weight the loss with the identity and let the Finsler structure act through the conditional paths and targets alone.

C. Amortized planning and execution

The pipeline is

$$\text{rollouts} \rightarrow \hat{\mathcal{E}} \rightarrow F_{\hat{\mathcal{E}}} \rightarrow v_\theta(\cdot | q_g, \hat{\mathcal{E}}) \rightarrow \text{plan or commands}, \quad (7)$$

shown in the strip across the top of Fig. 1. Velocity observations from command sweeps fit the envelope descriptor $\hat{\mathcal{E}}$ (the Randers or p -sum parameters, or in the drift-field experiments the wind strength), which defines $F_{\hat{\mathcal{E}}}$. Training pairs (q_0, q_1) are drawn independently from the start and goal distributions; each geodesic boundary value problem is solved by Gauss–Newton shooting on the initial momentum with continuation in the asymmetry and three initial branches, and a grid search with asymmetric edge costs certifies minimality, discarding pairs whose geodesic exceeds the grid time by more than 5% (10% on $\text{SE}(2)$). Obstacles enter as a conformal factor $s(q)F$, which preserves the Randers form; there the grid selects the homotopy class and the discrete Finsler length of the polyline is polished by gradient descent. Each geodesic contributes 48 samples (t, q_t, u_t) to a four-layer MLP of width 256 with SiLU activations whose inputs are t , the state $(x, y$ and $\cos \theta, \sin \theta$ on $\text{SE}(2)$), a goal code and $\hat{\mathcal{E}}$, trained with Adam for 6–8k steps at batch 2048.

At run time a plan is one RK4 integration of $\dot{q} = v_\theta(t, q | q_g, \hat{\mathcal{E}})$ from the current pose; executed on the envelope boundary its Finsler length is its travel time and its final velocity magnitude predicts the remaining time. For closed-loop execution we advance a flow clock $t_k = \tau_k / \hat{T}$ with the elapsed time τ_k and the initial estimate $\hat{T} = F(q_0, v_\theta(0, q_0))$, re-anchor it when the remaining time implied by the field magnitude deviates from the clock by more than 30%, and command the projected body velocity

$$\xi_{\text{cmd}} = \Pi_{\mathcal{B}(q)}(v_\theta(t_k, q_k)) = v_\theta / F(q_k, v_\theta), \quad (8)$$

so feasibility is guaranteed by construction and replanning costs one network evaluation per control step.

III. EXPERIMENTS

The experiments ask three questions: whether asymmetric geometry improves planning (Sec. III-B), whether the geometry can be learned and generalized (Sec. III-C), and whether the learned planner is practical (Sec. III-D).

A. Setup and baselines

All planners share the architecture, training pairs, goal conditioning and schedule of Sec. II-C and differ only in the conditional path family. *Euclidean* uses straight-line interpolation (optimal-transport conditional flow matching [21]). *Riemannian* uses geodesics of the symmetric part $\sqrt{v^\top A(q)v}$ of the same Finsler metric, the anisotropic but reversible geometry a Riemannian flow-matching policy [4] would use. *Finsler* uses the geodesics of Sec. II-B. The best-known reference is a grid search with asymmetric edge costs (32 directions at $h = 0.04$ in the plane, 98 directions at $h = 0.1$ and 10° on SE(2)); against geodesics verified by the boundary value solver it overestimates the true time by 0.4% in the plane (stable from $h = 0.08$ to 0.02) and by 2–3% on SE(2), so ratios below one lie within this band. We report three quantities. The *travel ratio* divides the executed travel time of the generated plan by the reference time from the start to the plan’s own endpoint, so it measures how efficiently a planner covered the ground it covered and cannot be improved by stopping short. The *time-to-goal ratio* instead divides by the reference time from the start to the goal, after adding the optimal remaining leg from the endpoint, and therefore charges every reach failure. We also report reach and collision rates and planning latency.

B. Does asymmetric geometry improve planning?

SE(2) pose planning. The body envelope is a Randers metric with forward 1.2, backward 0.5, lateral 0.4 m/s and yaw 1.2 rad/s, the asymmetry of a quadruped’s locomotion controller. The goal pose is the origin, start poses are uniform in $[-2.5, 2.5]^2 \times S^1$, and 192 held-out poses on a $5 \times 5 \times 8$ grid are evaluated; success requires a final position error below 0.2 m and a heading error below 15° . The symmetric baseline is the reversible ellipsoid with fore–aft speed 0.71 m/s, and the Euclidean baseline interpolates linearly in (x, y, θ) . Table I and Fig. 2 (top) show that the Finsler planner attains 0.979 of the grid reference time for every bearing of the goal. Because that reference is itself an upper bound, we also score the same plans against exact shooting geodesics on the 98 poses where the grid certifies the geodesic minimal, which removes the discretization bias rather than bounding it. Against the exact optimum the Finsler planner sits at 0.998, its residual accounted for by the 0.2 m position tolerance, while the symmetric and Euclidean baselines sit at 1.064 and 1.212. Optimality here is therefore verified, not inferred from a coarse reference. The Euclidean planner loses 30% when the goal is to the side, because it translates laterally at 0.4 m/s, and the symmetric planner loses 9–10% when the goal is behind or beside the robot, because it cannot distinguish backing up from walking forward and plans both at 0.71 m/s. The turn–walk–align maneuver that the Finsler flow discovers on its own is the qualitative signature of an asymmetric envelope.

These gaps belong to the conditional paths, not to the networks that learn them. Executed before any learning, the training paths themselves cost 1.205 of the optimal time for the Euclidean family and 1.058 for the symmetric Riemannian

TABLE I

SE(2) POSE PLANNING, 192 HELD-OUT POSES. EXECUTED TRAVEL TIME OVER THE REFERENCE TIME, SPLIT BY THE GOAL’S BEARING RELATIVE TO THE START HEADING; STANDARD ERRORS OF THE OVERALL MEANS ARE 0.013, 0.011 AND 0.003; EVERY PLANNER SUCCEEDS ON EVERY POSE; MEAN REFERENCE TIME 3.29 s.

Path	Ahead	Side	Behind	All	Worst
Euclidean	1.237	1.299	1.183	1.254	1.69
Riemannian	1.000	1.100	1.093	1.073	1.61
Finsler	0.960	0.985	0.986	0.979	1.25

family, against 1.000 for the Finsler geodesics; the learned fields add two to five per cent on top of those floors. The learner is correspondingly secondary to the geometry. Trained on the identical Finsler geodesics, and for four times as many steps as our flow, a trajectory regressor reaches 1.010 and a trajectory diffusion model in the style of Janner et al. [10] reaches 0.996, against 0.979 for the flow, which answers a query in 0.6 ms against the diffusion sampler’s 4.6 ms. The geometry buys the time and transfers to whatever learns it; what the flow adds is a field over the whole state–time domain rather than a single trajectory from a single start, so one model serves both as the planner and as the feedback law of Sec. III-D.

That parity is a property of this benchmark rather than of regression. Because the metric is left invariant, every SE(2) query reduces to the same goal and has one dominant homotopy class, so the last waypoint is a constant a regressor can memorize. On the planar drift field, where two goals are in play and the channel-core goal is approached by two symmetric branches, the same regressor trained on the same geodesics reaches 1.299 of the optimum with a worst case of 5.41, against 0.978 and 1.12 for the flow, because a deterministic map from start to waypoints must average branches that a flow keeps apart. A trajectory diffusion model reaches 1.045 there; its reach rate is not comparable to the other two, since Diffuser-style conditioning pins the final waypoint to the goal by inpainting whereas the flow and the regressor have to arrive on their own. The mechanism therefore matters exactly where the solution is multi-modal, and is close to irrelevant where it is not.

Spatially varying drift field. A Randers field on $\mathbb{R}^2$ models a head-wind channel $W(x) = (-\beta e^{-x_2^2/w^2}, 0)$, $w = 0.8$, whose core slows eastward motion to speed $1 - \beta$. Starts are uniform in a western box, goals are two blobs at the channel core (2.5, 0) and the channel edge (2.5, 1.6), and 98 held-out queries are evaluated at $\beta = 0.9$ (Fig. 2, bottom). The Finsler planner is time-optimal to within the reference error, with a mean travel-time ratio of 1.000 ± 0.001 , a median of 0.998 and a worst case of 1.02. Euclidean conditional paths cost $2.55\times$ on average and $6.3\times$ for core-to-core queries, and the Riemannian planner, which knows the anisotropy but not the direction of the wind, leaves an 8% gap with a worst case of 1.21. Charging each plan the optimal remaining leg from its endpoint, which accounts for the 0.89 reach rate of the curved-path planners at the slow core goal, gives time-to-goal ratios of

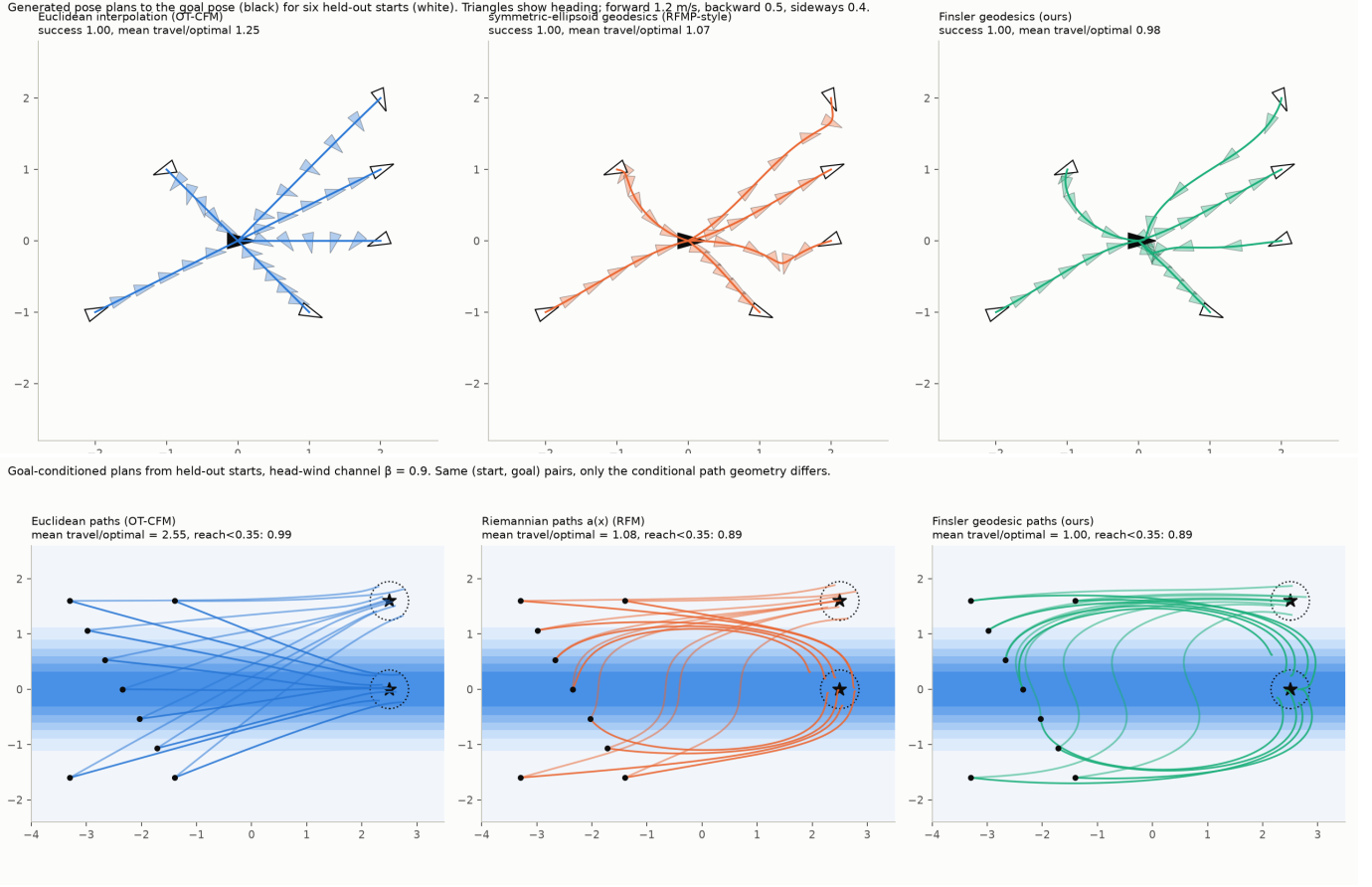


Fig. 2. Riemannian versus Finsler plans. Top: SE(2) plans to the origin pose from held-out start poses; Finsler plans turn towards the goal, walk forward and align the heading at the end, the symmetric planner backs up to goals behind the robot at the averaged fore-aft speed, and the Euclidean planner slides sideways. Bottom: plans in the head-wind channel ($\beta = 0.9$) for the same start-goal queries; Euclidean paths cross the channel core at speed 0.1, Riemannian paths detour but enter the core goal from the upwind side, and Finsler paths detour widely and enter the core goal downwind, reproducing the structure of Zermelo-optimal paths. Time-to-goal over optimal, averaged over 98 queries: Euclidean 2.58, Riemannian 1.15, Finsler 1.06.

2.58, 1.15 and 1.06. Weakening the wind to $\beta = 0.7$ shrinks the Euclidean loss to $1.47\times$ and leaves the Riemannian gap (1.06) and the Finsler optimality (1.00) unchanged.

A manipulator instance. The construction is not particular to legged robots. In the shoulder-elbow plane of a UR5, which carries the arm’s real link lengths, link masses and $120^\circ/\text{s}$ joint limit with the rated 5 kg payload, the achievable end-effector velocity set is $\{v : |u_i| \leq w_i, g(q)^\top u \leq P_{\max}\}$ with $u = J(q)^{-1}v$, so the gauge is $\max(|u_1|/w_1, |u_2|/w_2, \max(0, g(q)^\top u)/P_{\max})$. The power budget binds only when the motion consumes gravitational power, which makes the envelope non-reversible as a matter of physics rather than modeling. Gravity alone nevertheless leaves the routes untouched, as the supplementary analysis of closed drifts predicts: gravitational power is the derivative of a potential, the asymmetric part of the metric is therefore a closed one-form, and the symmetrized metric $\frac{1}{2}(F(v) + F(-v))$, which keeps the anisotropy and discards only the asymmetry, plans within 0.4% of the optimum. Power-and-force limiting adds an asymmetry that is not closed: capping the energy transferred on contact bounds the closing speed

by $\sqrt{2E_{\max}/m_{\text{eff}}(\hat{n})}$, with $m_{\text{eff}}(\hat{n}) = (\hat{n}^\top J M^{-1} J^\top \hat{n})^{-1}$ the effective mass along the approach direction, and leaves retreat unbounded; effective mass is quadratic in the direction and so reversible by itself, and the asymmetry lies entirely in the constraint being one sided. What that costs an arm is not what it costs a walking robot. Over four approaches to a workpiece the symmetrized metric picks routes within 1.0% of optimal, so route selection barely changes, while an isotropic metric loses 16.4%; but the symmetrized metric underestimates the travel time of *its own* plan by 45.5%, because the end effector rises at 0.51 m/s where it descends at 0.78. For a manipulator the geometry therefore buys a travel time that can be believed, which is what a cycle-time budget consumes, rather than a shorter route; where the asymmetry turns with the body or varies across space, as in the settings above, the route changes too.

Obstacles. Three discs enter the metric as a conformal factor $s(x)F$ with $s = 1 + 30 \sigma((r + 0.05 - d)/0.08)$, so all three planners are obstacle-aware and differ only in wind awareness (Fig. 3a). Over 98 queries at $\beta = 0.9$ the Finsler planner reaches 91% of goals with zero collisions at a median ratio

TABLE II
ENVELOPE IDENTIFICATION FROM NOISY ROLLOUTS ON SE(2). RADIAL
ERROR OF THE FITTED INDICATRIX AGAINST THE TRUE p -NORM
ENVELOPE, AND EXECUTED TIME OVER THE TRUE OPTIMUM FOR PLANS
MADE UNDER THE FITTED GEOMETRY (96 POSES).

Geometry	Err. mean	Err. max	Mean	Med.	Worst
Symmetric (Riem.)	15.1%	63%	1.126	1.059	1.74
Randers	9.0%	28%	1.036	1.031	1.14
p-sum Finsler	7.2%	19%	1.023	1.016	1.12

of 0.999 and a worst ratio of 1.05; the Riemannian planner, which routes around obstacles but not around the wind, reaches 71% and collides in 5% of queries at a median of 1.064 and a worst of 3.29, and the Euclidean planner triples the travel time (median 2.97).

C. Can the geometry be learned and generalized?

Envelopes from rollouts. The true envelope is an asymmetric p -norm box ($p = 4$; forward 1.2, backward 0.5, lateral 0.4 m/s, yaw 1.2 rad/s), whose flat faces place it outside the strongly convex families of Sec. II-A. We command 6000 body velocities, observe their projection onto the true envelope plus noise of standard deviation 0.02, and fit each family by least squares on the saturated observations with a hinge penalty on the interior ones (Fig. 4, bottom left). Plans computed under the fitted metric are executed under the true envelope from 96 start poses. Table II carries the central result: discarding the asymmetry costs 13% on average and 74% in the worst case, a one-term Randers fit with 9% mean radial error already plans within 3.6% of the true optimum, and the learned p -sum fit within 2.3%. Radial error converts into a much smaller planning loss because time-optimal paths use the forward and turning faces of the envelope, which every asymmetric fit captures.

Unseen asymmetry strengths. A single field $v_\theta(t, x | g, \beta)$ is trained on channel pairs whose wind strength is drawn from $\beta \sim U(0.30, 0.85)$, each pair with its own geodesic, and tested at an interior strength and at two strengths outside the training range (Fig. 3b). Finsler plans remain time-optimal at $\beta = 0.9$ and 0.95 (mean 0.999 and 1.000, worst 1.05 and 1.10), where the Euclidean loss grows to $2.5\times$ and $3.7\times$ and the Riemannian gap to 7–8%. The planner has learned the geometry as a function of the capability descriptor rather than the optimal field of one environment; we scope the claim as extrapolation over a one-parameter family of envelopes.

Execution on robots. We close the pipeline (7) on a Unitree H1-2 humanoid in MuJoCo [20], driven at 50 Hz by Unitree’s pretrained velocity-command locomotion policy [22]. Since that policy is trained on symmetric command ranges, we impose the asymmetric command limits of a deployment interface on it ($v_x \in [-0.5, 1.2]$, $v_y \in [-0.4, 0.4]$ m/s, $\omega \in [-1.2, 1.2]$ rad/s) and identify the envelope that results. A sweep of 80 command directions at 5 magnitudes (400 rollouts, 1 s settling and 5 s hold, achieved velocity averaged over the last 3 s, no falls) fits a Randers envelope with forward

1.20, backward 0.65, lateral 0.50 m/s and yaw 1.7 rad/s, whose symmetric part has fore–aft speed 0.84 m/s. Plans under the identified metric are executed as 20 Hz body-velocity commands at 90% of the envelope speed by the feedback rule (8), tracked by the locomotion policy (Fig. 4, top). Knowing the asymmetry makes the robot 29% faster to a goal 2.5 m behind it with reversed heading (4.8 against 6.7 s; it turns and walks instead of backing up), 23% faster to a goal 2 m beside it (4.0 against 5.2 s), and 22% faster to a goal 2.5 m ahead (3.0 against 3.8 s), where the symmetric planner walks at its averaged speed; no run falls. **We measure the envelope of the stock locomotion controller of a Unitree Go2 quadruped from motion-capture odometry under three conditions, nominal, a 5 kg payload and a conservative gait mode, obtaining forward, backward and lateral speeds of 1.5/0.6/0.5, 1.1/0.45/0.4 and 0.8/0.3/0.3 m/s and yaw rates of 1.5, 1.2 and 1.0 rad/s, and of a Unitree H1-2 humanoid, obtaining 1.1/0.45/0.35 m/s and 1.2 rad/s. Neither asymmetry is imposed; both are what the shipped controllers do. A single SE(2) field conditioned on the four envelope speeds, trained on geodesics of envelopes sampled around these values, plans for every condition and for both robots. Over twelve trials per task and condition the Finsler planner arrives 27–33% earlier than the symmetric planner on the quadruped (4.4 against 6.5 s nominal, 5.6 against 8.1 s with payload and 7.3 against 10.9 s in the conservative mode for the goal behind with reversed heading) and 24% earlier on the humanoid, the travel-time estimate of the open-loop plan is within 12% of the executed time, and no trial fails. Conditioning on capability is what makes this one model rather than five: attaching the payload during a run changes the descriptor, and the same field returns the new time-optimal plan at the next control step, where a grid planner would rebuild its graph in 9.9 s. Only the descriptor changes across conditions and across robots; the network, the feedback rule and the controllers are identical.**

D. Is the learned planner practical?

Latency. Fig. 3c compares the learned planner with the classical solvers that produced its training data, on the same CPU without a GPU. A full plan with 50 RK4 steps costs 13 ms in the plane and 29 ms on SE(2) (51 and 130 ms with 200 steps), a batch of one hundred queries costs 0.7 and 1.5 ms per query, and one evaluation of the feedback field costs $3 \mu\text{s}$. Grid search is competitive in the plane but on SE(2) needs a 10 s graph of 14 million edges plus 330 ms per new goal, with memory growing as h^{-d} , and the shooting solver costs 22 and 75 s per query.

The sharper point is what the query ranges over. A grid reference is built for one metric: change the envelope, because the robot picked up a payload or the terrain changed, and the graph must be rebuilt from scratch, which is the 0.14 s and 9.9 s of Fig. 3c rather than the per-goal sweep. The envelope-conditioned field answers a start, a goal *and* an envelope it has never seen in the same 13–29 ms, at the optimality reported in Sec. III-C for $\beta = 0.9$ and 0.95 outside its training range. Amortization over the capability descriptor is therefore

Goal-conditioned plans with obstacles (grey), head-wind channel $\beta = 0.9$. Obstacles are a conformal factor of each metric; only wind-awareness differs.

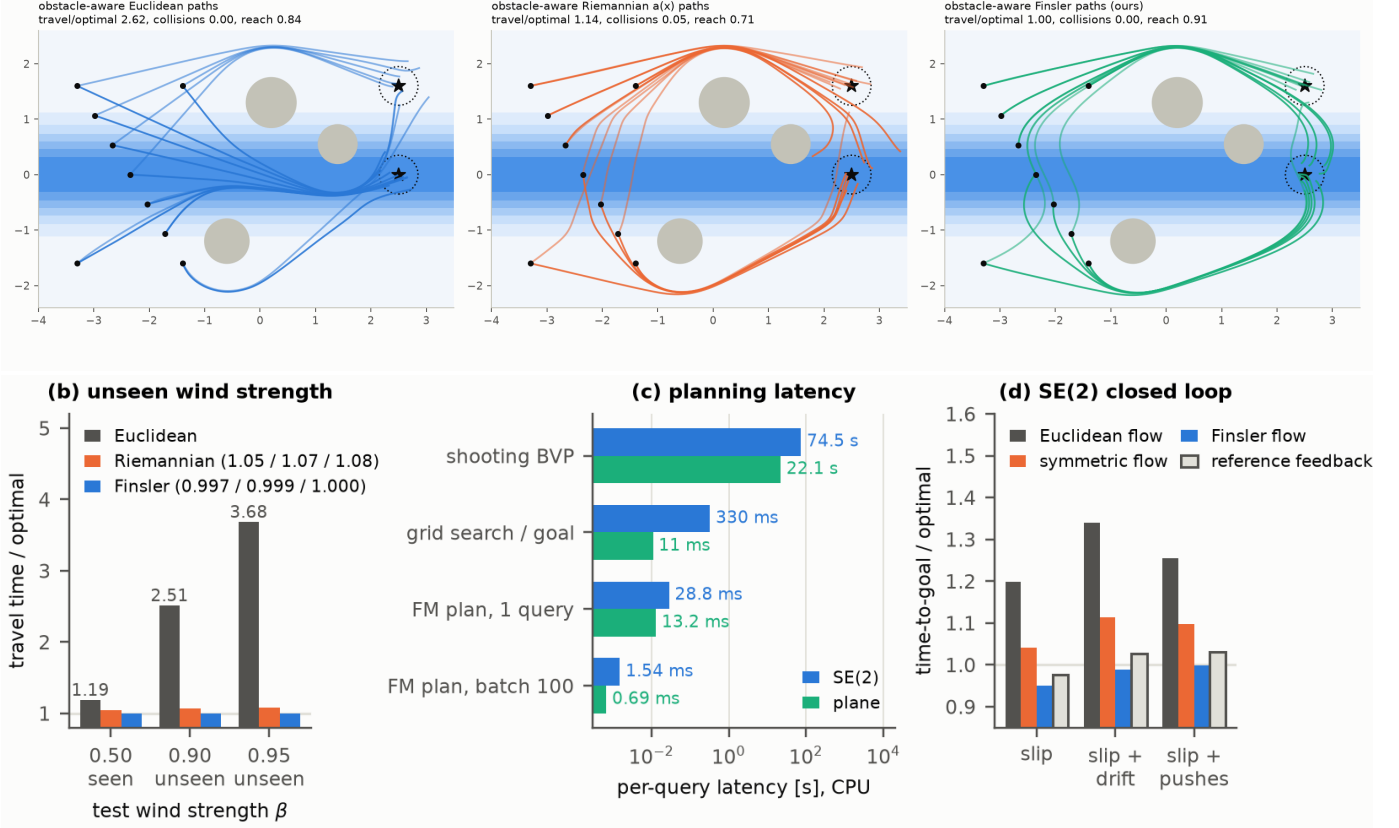


Fig. 3. (a) Plans with three obstacles in the head-wind channel; all planners are obstacle-aware, and the Finsler planner combines obstacle clearance with a downwind approach (reach 0.91, collisions 0, median ratio 0.999, worst 1.05). (b) One β -conditioned planner trained on $\beta \in [0.30, 0.85]$ and tested at unseen wind strengths. (c) Per-query latency of the learned planner against the solvers that produced its training data. (d) SE(2) closed-loop execution under disturbances; all methods reach every goal, and the Finsler feedback field matches the reference value-function descent.

a difference in kind rather than a constant factor, and it is the regime a robot whose envelope drifts during operation actually lives in.

Closed-loop disturbances. On SE(2) the flow at $t = 0$ is the initial velocity of the learned time-optimal path from the current pose; re-evaluated every 0.05 s and executed by (8) it is a feedback law. Fig. 3d reports 192 poses under slip noise, an unmodeled lateral drift of 0.15 m/s, and 0.3 m pushes every 2 s, against greedy descent of the reference value function. The Finsler field matches or beats the reference under every disturbance (0.951–0.998 of the undisturbed optimum) with success rate one, while the symmetric field loses 4–11% and the Euclidean field 20–34%. In the planar field, where the start distribution covers only part of the workspace, the streaming clock of Sec. II-C keeps the flow on its learned support: the streaming Finsler flow succeeds on every run under slip, $\pm 30\%$ wind gusts and pushes, at 0.97–1.01 of the optimal time. Finsler flow matching thus preserves the planning quality of the numerical solvers while amortizing the online computation, and one model covers both regimes: the field that generated the open-loop plan is the feedback law that executes it.

IV. LIMITATIONS

Our formulation assumes smooth, strongly convex velocity envelopes; strictly nonholonomic systems with zero lateral mobility call for a sub-Finsler extension. Planning quality inherits the accuracy of the identified envelope, and a deterministic flow commits to a single homotopy class per query where several are equally fast, which closed-loop replanning resolves at execution time. The present experiments address low-dimensional locomotion planning with envelope descriptors of a few parameters and fixed obstacle layouts; higher-dimensional configuration spaces, map-conditioned obstacles and envelopes that change online during execution are natural extensions of the same construction.

V. CONCLUSION

Robot capability envelopes are generally asymmetric and therefore cannot be represented faithfully by a reversible Riemannian geometry. Finsler flow matching converts these envelopes into conditional geodesic paths, with the target field taken from the forward distance through the Legendre transform and the loss weight kept independent of the sampled endpoint, and amortizes their solution into a fast planning

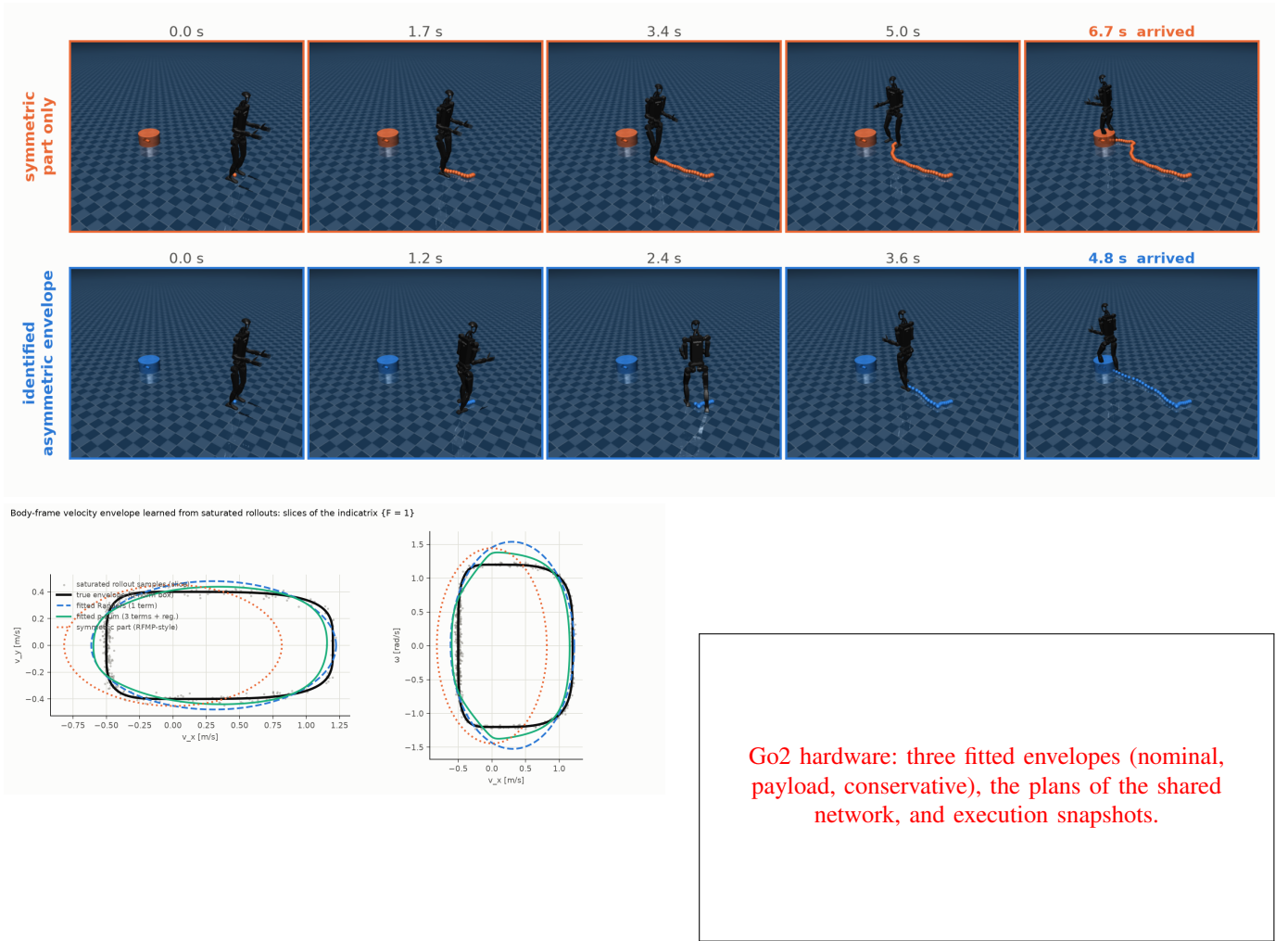


Fig. 4. Robot execution and learned envelopes. Top: the simulated H1-2 walking to a goal 2.5 m behind it with the heading reversed, under the two planners, with the goal pose and the path walked so far drawn into the scene. The symmetric planner cannot distinguish reversing from walking and swings out sideways; the planner that knows the identified envelope turns in place and walks straight in, arriving 29% earlier. Bottom left: sections of the true p -norm envelope and the indicatrices fitted from noisy rollouts; the symmetric fit averages forward and backward speeds and misses both. Bottom right: Go2 envelopes under three conditions and the corresponding executions.

flow. The resulting planner retains near-reference travel time across learned capabilities, obstacles, unseen asymmetry levels and closed-loop disturbances, where Euclidean and symmetric Riemannian planners lose up to $2.6\times$ and $1.13\times$. **One conditioned field covers a quadruped and a humanoid, and a capability that changes while the robot is moving, which is what a geometry built from the measured envelope buys over one that is designed.**

ACKNOWLEDGMENTS

Omitted for anonymous review.

REFERENCES

- [1] David Bao, Shiing-Shen Chern, and Zhongmin Shen. *An Introduction to Riemann–Finsler Geometry*, volume 200 of *Graduate Texts in Mathematics*. Springer, 2000.
- [2] David Bao, Colleen Robles, and Zhongmin Shen. Zermelo navigation on Riemannian manifolds. *Journal of Differential Geometry*, 66(3):377–435, 2004.
- [3] Hadi Beik-Mohammadi, Søren Hauberg, Georgios Arvanitidis, Gerhard Neumann, and Leonel Rozo. Learning Riemannian manifolds for geodesic motion skills. In *Robotics: Science and Systems (RSS)*, 2021.
- [4] Max Braun, Noémie Jaquier, Leonel Rozo, and Tamim Asfour. Riemannian flow matching policy for robot motion learning. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.
- [5] João Carvalho, An T. Le, Mark Baierl, Dorothea Koert, and Jan Peters. Motion planning diffusion: learning and planning of robot motions with diffusion models. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.
- [6] Ricky T. Q. Chen and Yaron Lipman. Flow matching on general geometries. In *International Conference on Learning Representations (ICLR)*, 2024.
- [7] Ching-An Cheng, Mustafa Mukadam, Jan Issac, Stan Birchfield, Dieter Fox, Byron Boots, and Nathan Ratliff.

- RMPflow: a computational graph for automatic motion policy generation. In *Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2018.
- [8] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin Burchfiel, and Shuran Song. Diffusion policy: visuomotor policy learning via action diffusion. In *Robotics: Science and Systems (RSS)*, 2023.
 - [9] Jonas Frey, Matias Mattamala, Nived Chebrolu, Cesar Cadena, Maurice Fallon, and Marco Hutter. Fast traversability estimation for wild visual navigation. In *Robotics: Science and Systems (RSS)*, 2023.
 - [10] Michael Janner, Yilun Du, Joshua B. Tenenbaum, and Sergey Levine. Planning with diffusion for flexible behavior synthesis. In *International Conference on Machine Learning (ICML)*, 2022.
 - [11] Sunshine Jiang, Xiaolin Fang, Nicholas Roy, Tomás Lozano-Pérez, Leslie Pack Kaelbling, and Siddharth Ancha. Streaming flow policy: simplifying diffusion/flow-matching policies by treating action trajectories as flow trajectories. In *Conference on Robot Learning (CoRL)*, 2025.
 - [12] Kacper Kapusniak, Peter Potaptchik, Teodora Reu, Leo Zhang, Alexander Tong, Michael Bronstein, Avishek Joey Bose, and Francesco Di Giovanni. Metric flow matching for smooth interpolations on the data manifold. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
 - [13] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023.
 - [14] Tapovan Lolla, Pierre F. J. Lermusiaux, Mattheus P. Ueckermann, and Patrick J. Haley. Time-optimal path planning in dynamic flows using level set equations: theory and schemes. *Ocean Dynamics*, 64(10):1373–1397, 2014.
 - [15] Gabriel B. Margolis and Pulkit Agrawal. Walk these ways: tuning robot control for generalization with multiplicity of behavior. In *Conference on Robot Learning (CoRL)*, 2023.
 - [16] Jean-Marie Mirebeau. Efficient fast marching with Finsler metrics. *Numerische Mathematik*, 126(3):515–557, 2014.
 - [17] Nathan D. Ratliff, Jan Issac, Daniel Kappler, Stan Birchfield, and Dieter Fox. Riemannian motion policies. *arXiv preprint arXiv:1801.02854*, 2018.
 - [18] Nathan D. Ratliff, Karl Van Wyk, Man Xie, Anqi Li, and Muhammad Asif Rana. Generalized nonlinear and Finsler geometry for robotics. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.
 - [19] Nikita Rudin, David Hoeller, Philipp Reist, and Marco Hutter. Learning to walk in minutes using massively parallel deep reinforcement learning. In *Conference on Robot Learning (CoRL)*, 2022.
 - [20] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: a physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012.
 - [21] Alexander Tong, Kilian Fatras, Nikolay Malkin, Guillaume Huguët, Yanlei Zhang, Jarrid Rector-Brooks, Guy Wolf, and Yoshua Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport. *Transactions on Machine Learning Research*, 2024.
 - [22] Unitree Robotics. unitree_rl_gym: reinforcement learning locomotion for Unitree robots. https://github.com/unitreerobotics/unitree_rl_gym, 2024.
 - [23] Tongzhou Wang, Antonio Torralba, Phillip Isola, and Amy Zhang. Optimal goal-reaching reinforcement learning via quasimetric learning. In *International Conference on Machine Learning (ICML)*, 2023.
 - [24] Lorenz Wellhausen, Alexey Dosovitskiy, René Ranftl, Krzysztof Walas, Cesar Cadena, and Marco Hutter. Where should I walk? Predicting terrain properties from images via self-supervised learning. *IEEE Robotics and Automation Letters*, 4(2):1509–1516, 2019.
 - [25] Ernst Zermelo. Über das navigationsproblem bei ruhender oder veränderlicher windverteilung. *Zeitschrift für Angewandte Mathematik und Mechanik*, 11(2):114–124, 1931.